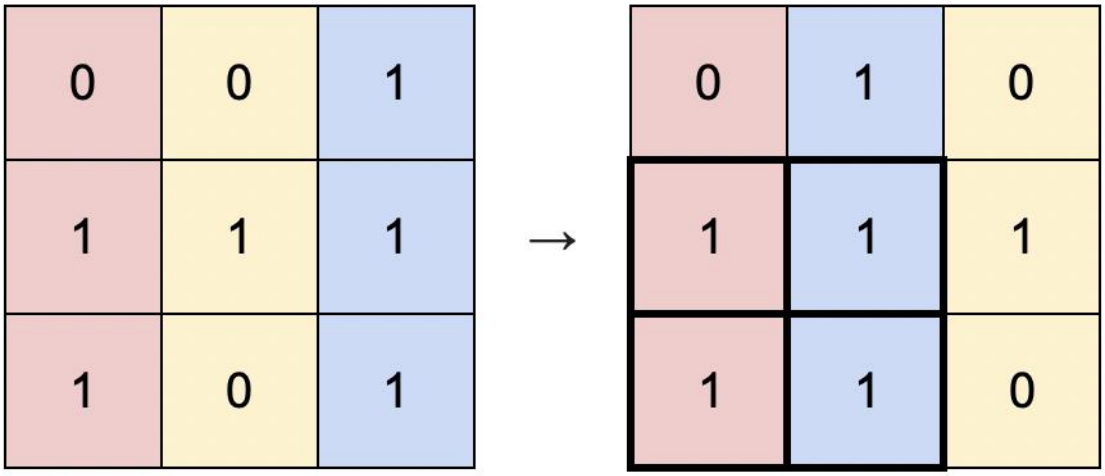
1727. Largest Submatrix With Rearrangements

You are given a binary matrix matrix of size m x n, and you are allowed to rearrange the **columns** of the matrix in any order.

Return *the area of the largest submatrix within* matrix *where* ***every*** *element of the submatrix is* 1 *after reordering the columns optimally.*

Example 1:



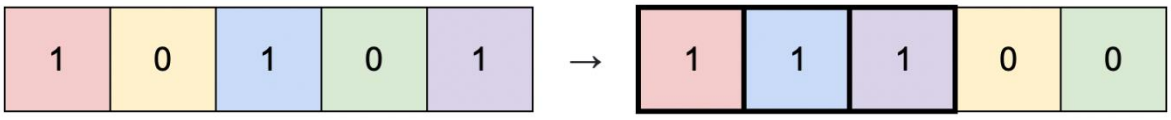
Input: matrix = [[0,0,1],[1,1,1],[1,0,1]]

Output: 4

Explanation: You can rearrange the columns as shown above.

The largest submatrix of 1s, in bold, has an area of 4.

Example 2:



Input: matrix = [[1,0,1,0,1]]

Output: 3

Explanation: You can rearrange the columns as shown above.

The largest submatrix of 1s, in bold, has an area of 3.

Example 3:

Input: matrix = [[1,1,0],[1,0,1]]

Output: 2

Explanation: Notice that you must rearrange entire columns, and there is no way to make a submatrix of 1s larger than an area of 2.

Constraints:

- m == matrix.length
- n == matrix[i].length
- 1 <= m * n <= 10^5
- matrix[i][j] is either 0 or 1.

Solution:--

```
class Solution {
    public int largestSubmatrix(int[][] matrix) {
        int m = matrix.length;
        int n = matrix[0].length;
        int ans = 0;

        for (int row = 0; row < m; row++) {
            for (int col = 0; col < n; col++) {
                if (matrix[row][col] != 0 && row > 0) {
                    matrix[row][col] += matrix[row - 1][col];
                }
```

```
        }

        int[] currRow = matrix[row].clone();
        Arrays.sort(currRow);
        for (int i = 0; i < n; i++) {
            ans = Math.max(ans, currRow[i] * (n - i));
        }
    }

    return ans;
  }
}
```